# Applying IEC 60909, Fault Current Calculations

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***Abstract*—Rather than the short-circuit current that would occur in a specific instance, International Electrotechnical Commission 60909 derives the maximum and minimum prospective short-circuit currents in a system for each specific location and time. This is reported using a series of parameters which relate to the rated short-circuit current of equipment and the tests required on equipment to prove that rating. The influence of arc voltage on short-circuit currents is then discussed.**

***Index Terms*—Arc voltage, balanced, calculations, dc component, impedance, Joule integral, maximum, minimum, peak, rating, short-circuit current, symmetrical component, system voltage, testing, unbalanced, $X/R$.**

## I. Introduction

A SHORT-CIRCUIT current is the result of an unwanted event on a power system that needs to be managed without causing extensive damage. The protection system must clear the short-circuit current, and all the equipment subject to the short-circuit current must not be damaged by it.

In order to achieve this outcome, equipment specifications require testing with short-circuit currents defined by specific parameters.

The equipment, however, is unlikely to ever see a short circuit with the particular parameters specified in its equipment specification. Even a test station is unlikely to match the parameters exactly.

The applications engineer needs to establish whether the guaranteed short-circuit current capability of the equipment is likely to be exceeded by any of the events that could occur in the system where the equipment is or will be deployed.

This requires calculation of the short-circuit currents that may occur under all the normal operating conditions of the system. This is different to calculating the short-circuit current that has or will occur in a very specific case where every piece of data is accurately known.

Each short-circuit current has a time-varying waveform which needs to be reduced to a set of parameters so that it can be specified for equipment testing and reported in system calculations.

The short-circuit current parameters that are used relate to how the short-circuit current affects the equipment that must carry or interrupt it.

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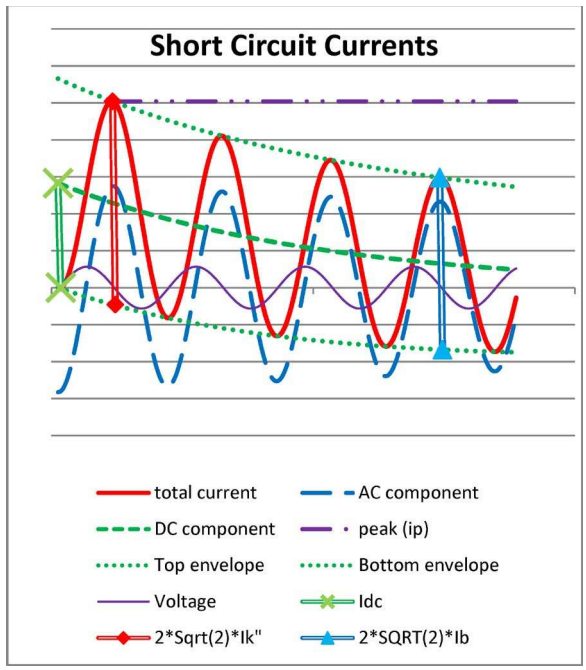



Fig. 1. Typical waveforms: (Solid) Total short-circuit current, (dashed) ac component (decaying if near to generator), and (dotted) transient dc component with (dotted) top and bottom envelopes. IEC 60909 parameters: $i_p$ (peak), $I_{\mathrm{dc}}$ (dc component), $I_k{''}$ (initial symmetrical component) times $2\sqrt{2}$, and $I_b$ (typical breaking current) times $2\sqrt{2}$.

## II. Short-Circuit Current Components

In a simple $R + jX$ inductive circuit, a short-circuit current consists of a decaying ac component and a decaying dc component. These components add together to provide the total current.

Close to a generator, the generator's change in reactance with time causes the ac current to fall with time. Far from a generator in the supply grid, the ac component is constant.

The dc component of the short-circuit current is due to the fact that current will not change instantaneously in an inductance. At the instant that the short circuit occurs, the ac component will have the same amplitude and phase angle that it would have had if it had been there for some time (excluding ac decay).

Since the instantaneous ac value is rarely zero, a decaying dc transient is generated with the opposite amplitude to the ac value at the start of the short-circuit period. This allows the short-circuit current to start from the instantaneous value of current prior to the short circuit.

This produces the current waveforms shown in Fig. 1 for the case of maximum dc component and asymmetry in the total current waveform. On a highly inductive power system, the

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maximum asymmetry is produced by a short circuit that begins just before a voltage zero.

In order to produce this waveform from equations, it is necessary to specify the voltage amplitude, frequency, $X/R$ ratio, and impedance of the system plus the decay constant for the ac waveform. The decay of any dc component is set by the $X/R$ ratio.

These resultant ac and dc components then must be added together to produce the total waveform and the peak.

Unfortunately, in real power systems, the reactance near to a generator is a function of time. Also, in power system components, the resistance is a function of frequency due to the skin and proximity effects interacting with the large physical dimensions.

While the equation parameters can plot simple system responses, large complex systems cannot always be easily reduced to simple equations with only a few parameters. The results of complex calculations and tests need to be easily measured and reported in a manner that allows system calculations to be compared with test data in a manner that correlates with how the short-circuit currents react with equipment.

## III. IEC 60909 Short-Circuit Current Parameters

IEC 60909 describes how to calculate and measure the resultant short-circuit current waveform at a specified location in a network and at a specific time, either the point of initiation of the short circuit or contact opening of switchgear.

1) A **prospective (available) short-circuit current** is the current that would flow if the short circuit were replaced by an ideal connection of negligible impedance without any change of the supply.

   Note: This is not the current that would flow during a short-circuit test on equipment that has impedance or produces a voltage but the current that would flow in a bolted fault at its incoming terminals. This allows for the equipment to change the current to help it survive. (For example, fuses and current-limiting circuit breakers are specified using a prospective current that never flows during a test due to the arc voltage limiting the current.)
2) The **initial symmetrical short-circuit current $\mathbf{I_k}''$** is the rms value of the ac symmetrical component of a prospective (available) short-circuit current applicable at the instant of the short circuit.

   Note: This is an rms value of only the ac component not the total waveform and is measured between the top and bottom envelopes of the current waveform. It is defined at the instant of short circuit because the ac component can decay for near to generator faults and a defining value is required.
3) The **decaying (aperiodic) component $\mathbf{i_{dc}}$ of a short-circuit current** is the mean value between the top and bottom envelopes at the time the short circuit starts.

   Note: This describes how to measure the value from a measured or calculated waveform.
4) The **peak short-circuit current $\mathbf{i_p}$** is the maximum possible instantaneous value of the prospective (available) short-circuit current.

   Note: This is not the peak value for a particular event but the highest value depending on $X/R$ and the phase angle when the short circuit starts.
5) The **symmetrical short-circuit breaking current $\mathbf{I_b}$** is the rms value of an integral cycle of the symmetrical ac component of the prospective short-circuit current at the instant of contact separation of the first pole to open of a switching device.

   Note: This defines how to measure the rms value of the ac component at the point in time that is relevant to switching devices. This only differs from $I_k''$ near to generators where the ac component falls with time.
6) The **steady-state short-circuit current $\mathbf{I_k}$** is the rms value of the short-circuit current after the decay of transient phenomena.

   Within the International Electrotechnical Commission (IEC) system, these five parameters are used to define all short-circuit tests and the calculation of the potential short-circuit currents in power systems.

## IV. Other Short-Circuit Current Parameters

Outside of the IEC sphere of influence, other parameters will be found that describe short-circuit currents.

1) The **asymmetrical rms short-circuit current is** often defined as the rms value of the first half cycle of an asymmetric current.

Note: This is calculated from initiation until the current changes sign and can be 65% larger than the symmetrical rms current (ac component) of the same waveform in highly reactive circuits where it lasts a similar percentage longer.

## V. Short-Circuit Carrying Capacity

The short-circuit strength of power system components can be specified in a number of ways:

1) rated short-time withstand current $I_{\mathrm{rms}}$ together with rate peak $i_p$ withstand current;
2) rated prospective short-circuit withstand current $I_{\mathrm{rms}}$;
3) rated conditional short-circuit current $I_{\mathrm{rms}}$;
4) rated fused short-circuit current $I_{\mathrm{rms}}$.

The specified $I_{\mathrm{rms}}$ current must occur for at least 0.1 s and the specified $i_p$ achieved during through-fault tests for withstand current ratings in many equipment standards.

For prospective current ratings, the specified $I_{\mathrm{rms}}$ occurs at the input terminals of the device, and the actual test current can be smaller due to the natural impedance of the device (busbar or switchboard).

For the conditional and fused ratings, the protection specified is allowed to limit the actual current during testing.

The value of $i_p$ determines the peak mechanical forces in equipment and is therefore a major parameter in failure mechanisms.

It is common, however, that the test station cannot achieve both the specified $I_{\mathrm{rms}}$ and the corresponding value of $i_p$ required by the equipment standard at the same time. This is because the $X/R$ ratio assumed in the equipment standard and the $X/R$ ratio of the test station circuit rarely match.

Test stations therefore commonly perform a peak test of a few cycles followed by a thermal test.

$I_{\rm rms}$ needs to be achieved at the start of the short circuit for a prospective rating and for up to 0.1 s for a withstand rating. Remember that the ac component decays in generator test stations.

The Joule integral or $I^2t$ of the test also needs to exceed the rated $I^2_{\rm rms}$ times the rated time, which may be specified or come from the equipment standard. The Joule integral of the current $\int i^2 \cdot dt$ is a measure of the heat dissipated in the resistances of the system and therefore determines the temperature reached by the components.

The rated short-circuit current $I_{\rm rms}$ therefore defines three parameters of a piece of equipment, which need to be satisfied by test.

1) $I_{\rm rms}(\text{required}) > I_{\rm rms}(\text{specified})$.
2) $i_p(\text{required}) > i_p(\text{specified})$.
3) $I^2_{\rm rms}T(\text{required}) > I^2_{\rm rms}T(\text{specified})$.

## VI. Comparing Specified Capability With System Requirements

While the equipment capability is proven by test, the system requirements need to be calculated in accordance with a standard that relates to the tested equipment capability.

On a new power system, the rated $I_{\rm rms}$ of the equipment is normally selected from the R10 range of numbers with a reasonable margin above the calculated ${I_k}''$ or prospective short-circuit current of the system.

On an existing power system where the calculated short-circuit current is approaching the ratings of some of the equipment, much more care is required. All three tests need to be applied to prevent equipment failure.

1) $I_{\rm rms}(\text{rated}) > I_{\rm rms}(\text{calculated}) = {I_k}''$.
2) $i_p(\text{rated}) > i_p(\text{calculated}) = i_p$.
3) $I^2_{\rm rms}T(\text{rated}) > I^2_{\rm rms}T(\text{calculated}) = {I_k}''^2T_k$.

The peaking factor in equipment standards (such as IEC 61439 [2] and IEC 62271 [3] for switchgear and controlgear) may be lower than that in the system due to the assumed $X/R$ ratio, so a simple $I_{\rm rms}$ test is not sufficient.

If reclosers are used, the heat generated cannot dissipate in the time between recloses so the sum of all the reclose durations is required.

To compare with the equipment rated values, IEC 60909 calculates ${I_k}''$, the initial rms value of the symmetrical component, $i_p$, the maximum value of the peak current, and $I^2_{\rm rms}T_k$ where $T_k$ is the sum of the durations of each short-circuit current.

If any of the rated values are conditional, the applications engineer must ensure that the conditions (protection required) comply with what was used during the certification testing.

If American equipment is being used, the applications engineer must ensure that he does not compare the asymmetrical fault-current rating of a piece of equipment with a calculated symmetrical short-circuit current because these are two different things with significantly different values. The asymmetrical value includes the two IEC 60909 components of symmetrical ac and decaying dc components into one combined value, which can be 65% larger than the value of the ac component.

## VII. Switching Equipment

While switching equipment must be capable of carrying a through-fault current when it is not called upon to close or trip, there are many more issues to consider than the fault current withstand requirements for cables, overhead lines, transformers, busbars, and the busbar systems of switchboards.

Except for the asymmetrical breaking current test (test duty 5), all the breaking current tests are carried out with symmetrical currents, most of which are directly related to $I_{\rm rms}$, the rated symmetrical current of the device.

This defines one of the most significant parameters of current extinction $di/dt_{i0}$ or the rate of change of current at current zero. Not only does this defines the rate of contraction of the arcing column and the rate of rise of voltage withstand of the extinction mechanism after current zero, it also sets the rate of rise of the voltage applied by the circuit.

It is therefore important, at the instant of contact separation, to have the rated symmetrical breaking current of the switchgear greater than $I_b$, the symmetrical ac component of the prospective short-circuit current.

The peak current $i_p$ affects the switchgear in a number of ways. During a making test, after the contacts prestrike, $i_p$ defines the maximum pressure that the mechanism must overcome to force the contacts to metallic closure (i.e., mechanism strength). During the asymmetrical opening tests, it sets the maximum pressure in extinction chambers. The following three tests are therefore equally important in establishing the suitability of switchgear.

1) $I_{\rm rms}(\text{rated}) > I_{\rm rms}(\text{calculated}) = {I_k}''$.
2) $i_p(\text{rated}) > i_p(\text{calculated}) = i_p$.
3) $I^2_{\rm rms}T(\text{rated}) > I^2_{\rm rms}T(\text{calculated}) = {I_k}''^2T_k$.

They are not the only criteria however. Some switchgear suffers from minimum and/or critical breaking current issues.

High-voltage back up fuses will explode if subjected to currents below their minimum breaking current. (When not all of the notches in the parallel elements initially clear, this leads to restrikes inside the fuse cartridge, excessive energy dissipation, and eventual rupture.)

Oil circuit breakers have critical breaking currents that require special tests that need to be monitored.

## VIII. Unbalanced Short Circuits

While three-phase equipment normally has only one rated short-circuit current, it relates not only to three-phase faults but two-phase and single-phase faults as well, and the equipment standards often set out extra tests to cover the different types of fault current.

As well as the initial symmetrical (three phase) short-circuit current, IEC 60909 sets out how to calculate the following:

1) line-to-earth short-circuit currents ${I_{k1}}''$;
2) line-to-line short-circuit currents ${I_{k2}}''$;
3) line-to-line-to-earth short-circuit currents:
   a) ${I_{k2\rm EL2}}''$;

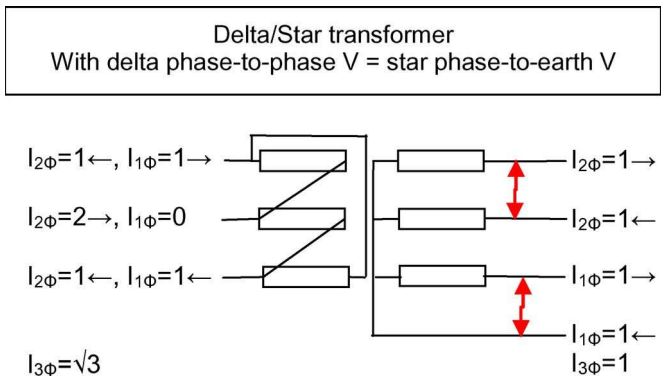



Fig. 2. Transformation of unbalanced fault currents across a delta star transformer.

b) $I_{k2\mathrm{EL3}}{}''$;
c) $I_{k\mathrm{E2E}}{}''$.

Note: In line-to-line-to-earth short circuits, the currents in each phase and earth all have different amplitudes.

As well as the different currents in the fault, each of these unbalanced faults has different current transformation ratios across transformers depending on the vector group of the transformer (Fig. 2).

While with the common delta/star vector group a three-phase current transforms with the ratio of the high-voltage to low-voltage phase-to-phase voltages, a single-phase current transforms with the ratio of the high-voltage phase-to-phase voltage to the low-voltage phase-to-earth voltage in two of the three high-voltage lines.

With unganged high-voltage fuses, after the first fuse clears in a three-phase low-voltage fault, this leads to a reduction of current in the remaining high-voltage fuses. There has been at least one case where the second and third fuses never cleared, and manual tripping was required hours later.

With a delta/star transformer, a low-voltage phase–phase current transformed to high-voltage line currents becomes, in one phase, two and, in two phases, one times the high-voltage phase-to-phase voltage divided by the low-voltage phase-to-earth voltage.

The different transformation ratios for different faults can lead to grading issues.

## IX. IEC 60909 Calculations

The IEC 60909 calculation method uses an equivalent source voltage at the short-circuit location driving into the short-circuit impedance of the network with all other voltage sources set to zero.

Symmetrical components are used to define the positive sequence, negative sequence, and zero sequence impedances of the system and its components in order to calculate the unbalanced short-circuit currents.

Any power system with multiple voltage levels and voltage control using transformer tap changers and power factor control has many different configurations per day let alone per year.

The standard sets out how to derive what is the most likely prospective maximum and minimum fault currents at that location even though tap ratios, loads, and power factor correction are continually changing.

This involves procedures for deriving the short-circuit impedance of the various system components including impedance correction factors, which ensure that results both near equipment and out in the network represent the most probable outcomes.

This allows transformer impedance to be calculated in the main tap-changer position and shunt capacitance and nonrotating loads to be neglected.

This reduces a million different calculations to one maximum and one minimum.

It should be noted that most of the data required are often not available and default values are required. These are provided in the different parts of IEC 60909.

## X. Maximum and Minimum Short-Circuit Currents

Maximum short-circuit currents need to be calculated because they determine the rating required for the equipment on the system. They should allow for foreseeable system upgrades that could occur independently of repeating the calculations and confirming ratings.

While IEC 60909 does not call for it, it is useful to also calculate a present maximum, which is needed to check present protection grading and investigate incidents.

Calculation of minimum short-circuit currents is required to ensure that protection will pick up and trip with the setting applied. The protection also must not explode because the current is below the minimum breaking current of some equipment.

IEC 60909 introduces the concept of a voltage factor "c" to calculate the maximum and minimum short-circuit currents.

For the maximum short-circuit current, $c_{\max}U_n$ is equal to the highest system voltage $U_m$ from IEC 60038 (standard voltages) or the highest voltage for equipment and represents the highest voltage that should appear under normal conditions at any location in the network at that voltage level.

For the minimum short-circuit current, $c_{\min}U_n$ is the lowest supply or utilization voltage from IEC 60038 at that location in the network.

The two voltage factors should be used with the most likely impedance of each component. If accurate data are not available, the most likely value rather than an extreme value should be used. In other words be conservative once only and not with the choice of every parameter.

The impedance correction factors are derived for the maximum case and need to be applied with thought for the minimum cases.

The supply network is not the same for the maximum and minimum cases.

For the maximum case, the following data should be included.

1) $c_{\max}U_n$.
2) The impedance corresponding to the maximum short-circuit current that the supply authority reserves the right to provide to the premises. (New transformer just outside.)
3) All parallel transformers in service.

4) If there is a ring main, it should be closed.
5) Calculation performed at input terminals of a low-voltage switchboard.

For the minimum case, the following data should be included.

1) $c_{\min}U_n$.
2) The impedance corresponding to the minimum short-circuit current that the supply authority guarantees to provide that connection. (Together with voltage control this can be derived from minimum capacity.)
3) Minimum parallel transformers in service (at least one out of service if it can run that way).
4) If there is a ring main, it should be open.
5) Transformer to switchboard impedance included.
6) Calculation performed at the end of a low-voltage switchboard, where an arc will run to and burn.
7) A calculation including the influence of the arc voltage for low voltage.

## XI. Influence of Arc Voltage on Short-Circuit Current

The impedance of an electrical supply network is mainly inductive with a small resistive component, and the impedance of loads is mainly resistive with an inductive component (0.8 power factor). This keeps the system losses within reasonable levels.

The voltage across an arc is however in phase with the current and out of phase with the reactive-impedance voltage drop.

While arcs on parallel electrodes tend to produce a sine-wave-like voltage waveform (with plenty of distortion), they are not resistive because of the negative $v/i$ characteristic. (From zero to around 1000 A (instantaneous), the arc voltage/cm falls to around 10 V/cm and thereafter rises slowly with current. This dominates behavior which is different to a resistor where the voltage increases with instantaneous current.)

The arc voltage is also only a weak function of electrode separation and therefore system voltage. The sine-wave-like waveform is due to the arc lengthening and shortening during each cycle, and the same behavior dominates over the influence of electrode separation.

The following vector diagram shows the influence of a 350-$V_{\rm rms}$ equivalent arc voltage (fundamental frequency component phase to phase) on a 20 000-A bolted short circuit in a nominal 460-V system with an $X/R$ of ten (Fig. 3). It was calculated at the utilization voltage minimum of 424 V.

The 20-kA prospective fault current at the maximum 508 V drops to 17.3 kA at the minimum 424 V. With 350-V arcing volts, the current falls to 8.2 kA or 41% of the maximum prospective current and 47% of the minimum nonarcing short-circuit current.

Even on a nominal 600-V system, the arc voltage has significant influence (Fig. 4). The 20-kA prospective fault current at the maximum 635 V drops to 16.4 kA at the minimum 520 V. With 350-V equivalent arcing volt phase–phase, the minimum current falls to 11.2 kA or 56% of the maximum prospective current and 68% of the minimum nonarcing current.

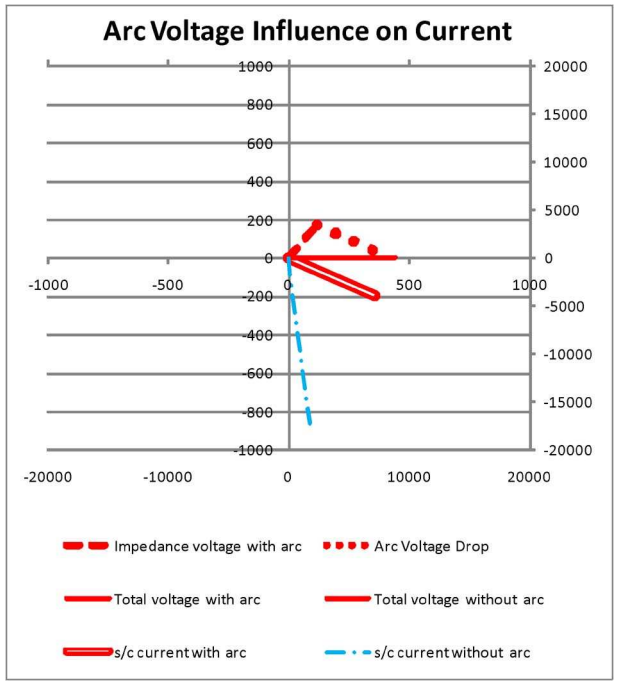



Fig. 3. Vector diagrams of a nominal 460-V 20-kA bolted fault and the same system with a 350-V equivalent arc voltage. Scales are voltage (1000 V) and current (20 000 A).

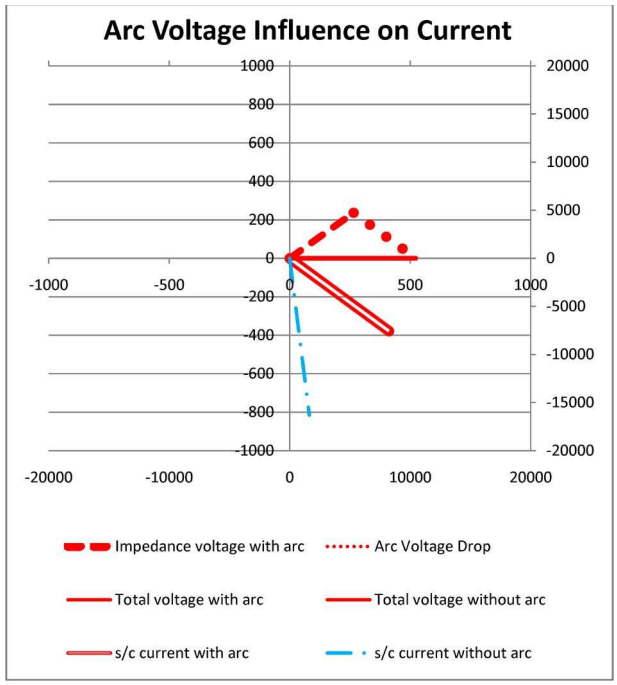



Fig. 4. Vector diagrams of a nominal 600-V 20-kA bolted fault and the same system with a 350-V equivalent arc voltage. Scales are voltage (1000 V) and current (20 000 A).

It is a completely different story for a 20-kA bolted short circuit at the maximum 4.44 kV in a nominal 4.16-kV system with the same $X/R$ of ten (Fig. 5). The minimum voltage of 3.6 kV drops the short-circuit current to 16.2 kA.

The 350-V equivalent phase–phase arcing voltage only reduces the 16.2 kA to 16 kA or 99% of that when arcing voltage is not considered. This is because not only the arc voltage is ten

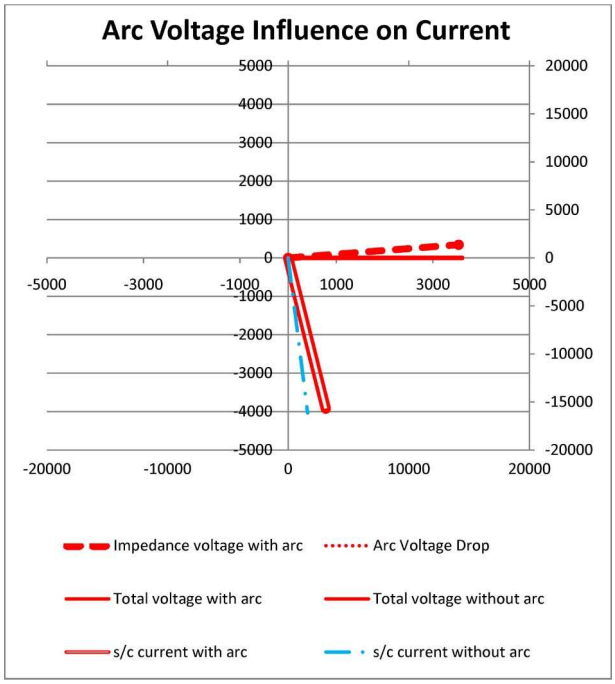



Fig. 5. Vector diagrams of a nominal 4.16-kV 20-kA bolted fault and the same system with a 350-V equivalent arc voltage. Scales are voltage (5000 V) and current (20 000 A).

times smaller in relation to the system voltage but also it is at 79° to it, and therefore, it has an even smaller influence.

## XII. Arcing Hazards

With the normal time-graded protection on distribution systems, the largest arc energy is normally produced by the minimum arcing fault current because the influence of the increased time for the protection to trip exceeds the influence of the smaller arcing current. This is the reason that minimum short-circuit currents including the arc voltage influence are an essential calculation.

In some systems, the minimum short-circuit current during arcing may not trip the protection at all.

This is not only a case of insufficient current to pick up the protection. The waveform of low-voltage fault currents can include periods where all three currents remain near zero for a number of cycles before the insulation fails again. The protection must be able to detect such discontinuous waveforms without resetting.

## XIII. Conclusion

IEC 60909, short-circuit currents in three-phase ac systems, calculates the prospective maximum and minimum values of the short-circuit current at a particular location in a power system independent of the continually varying load currents and transformer tap positions but taking into account variations in system configurations.

The results of IEC 60909 calculations are reported as follows:

1) the rms value of the symmetrical component of the short-circuit current:
   a) $I_k''$ at the start of the short circuit;
   b) $I_b$ at contact opening of protection;
2) the maximum peak current $i_p$;
3) the Joule integral of the current ${I_k''}^2 T_k$.

It has been shown how these parameters relate to the rated short-circuit current $I_{\rm rms}$ of equipment and how short-circuit tests on equipment, in turn, relate to its rated short-circuit current.

As well as balanced three-phase short-circuit currents, IEC 60909 provides procedures to calculate unbalanced currents $({I_{k1}}'', {I_{k2}}'', {I_{k2\rm EL2}}'', {I_{k2\rm EL3}}'',$ and ${I_{kE2E}}'')$, which are particularly important in setting protection.

While IEC 60909 does not take into account arcing voltage, it has been shown that, while arcing voltage is a minor issue at high voltage, it has a considerable impact at low voltage.

## References

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He spent four years in Switzerland with Brown Boveri where he worked on circuit breaker development and testing before returning to Australia to join the Sydney County Council, an electrical distribution authority where he held positions in substations and overhead transmission and ran the high-power testing station. In 1980, he joined Bassett Consulting Engineers where he became an Associate Director. In 1989, he formed Sweeting Consulting, St. Ives, Australia, which is a specialist high-voltage electrical engineering firm. He is a Visiting Professor at the University of Wollongong, Wollongong, Australia. He has published 70 papers on circuit breakers, electric arcs, testing, switching over voltages, cable fault location, motor failures, computer-aided design, standards, safety, standard voltages, power quality, and electric arc burn hazards.

Dr. Sweeting is an Honorary Fellow of The Institution of Engineers Australia, a Senior Member of the IEEE Industry Applications Society, and a member of Consult Australia and the International Council on Large Electric Systems (CIGRE). He has been a member of a number of International CIGRE committees, International Electrotechnical Commission working groups, and Standards Australia's committees covering short-circuit currents and power quality.